

ECON100B - SECTION 9

The Romer Model

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ANNOUNCEMENTS/AGENDA

Announcements

- PSET 2 due Friday, 2/27
- In-Class Midterm, 3/5

Agenda

- Romer Model Wrap Up
- Practice and PSET

Question of the Day

PSET 2 Q1.c: Consumption goods and services include things like food and health care. Investment goods and services include container cranes for seaports, architectural services, and the office buildings themselves. Do the same workers in the economy typically produce both consumption goods and investment goods? Explain.

PSET 2 Q1.d: Given your previous answers, what should be true in the short run about the U.S. share of labor engaged in producing consumption goods, if consumption and investment sectors require and benefit from the same amount of capital and technology?

What is your personal “Roman Empire”? What is something that lives rent free in your head?

The Romer Model



The Romer Model Setup

- For now, let's ignore capital and just focus on Ideas and Labor

$$Y_t = A_t L_{yt}$$
$$\Delta A_t = \bar{z} A_t L_{at}$$

- Output is produced using the stock of existing knowledge and labor
- The change in the stock of ideas is the “number of new ideas” produced in time period t .
- Ideas are produced using existing ideas and workers, multiplied by a parameter “ z ”, which represents how good our economy is at producing ideas. Workers can either produce output (y) or ideas (a)

$$L_{yt} + L_{at} = \bar{N}.$$

The Romer Model – Setup

- How do we determine how much labor goes toward innovation vs. output?
- In this class, we're given the parameter exogenously:

$$L_{at} = \bar{l}\bar{N}$$

- Food for thought: In a market, the market will naturally try to find the optimal point of innovation. What would happen to our parameter $\bar{l}$ under an unregulated free market economy? How about an economy that has patents and protections for innovation?

Solving the Romer Model

- Output per person can be represented as:

$$y_t \equiv \frac{Y_t}{\bar{N}} = A_t(1 - \bar{\ell}).$$

- We can see that output per person depends on the total stock of knowledge (nonrivalry) as opposed to per-capital stock as we saw in the Solow model
- We can find our growth in ideas with the following:

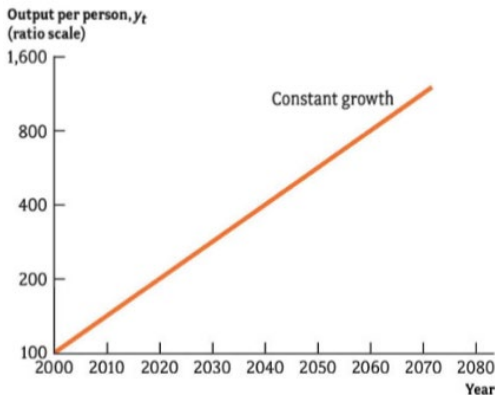
$$\frac{\Delta A_t}{A_t} = \bar{z}L_{at} = \bar{z}\bar{\ell}\bar{N}.$$

$$A_t = \bar{A}_0(1 + \bar{g})^t.$$

Solving the Romer Model

Our final Romer model output per person equation is:

$$y_t = \bar{A}_0(1 - \bar{l})(1 + \bar{g})^t$$



Practice with the Romer Model

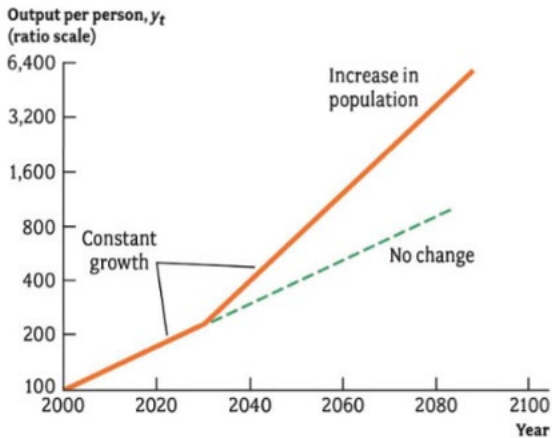
1. What would happen to the growth rate of output per person if we increased the number of people in the population (N)? What does this tell us?
2. What would happen to the growth rate of output per person if we increased the share of people that were working in research (l)? What does this tell us?
3. Suppose we're given a Romer model with the following parameters:
 $\bar{z} = 0.01, \bar{l} = 0.05, \bar{N} = 100, \bar{A}_0 = 1000$

What is our output per person in 10 years? What is our output per person in 20 years? In 100 years?

Recall: $y_t = \bar{A}_0(1 - \bar{l})(1 + \bar{g})^t$ and $\bar{g} = \bar{z}\bar{l}\bar{N}$

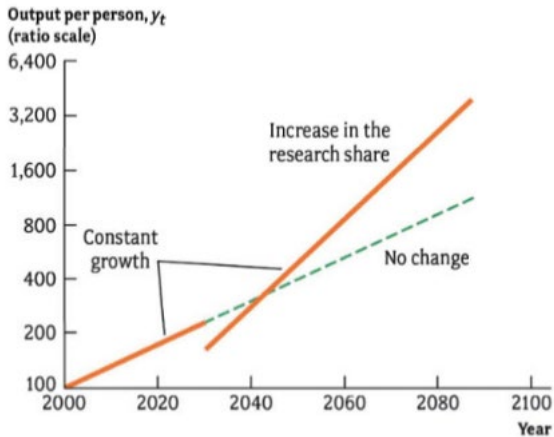
Practice Solutions

Output per Person after an Increase in $\bar{N}$



Practice Solutions

Output per Person after an Increase in ℓ



Practice Solutions

3. First calculate the growth rate g : $0.01 * 0.05 * 100 = 0.05$

Then plug into our output per person formula to find:

10 year output per person: 1,547

20 year output per person: 2,520.63

100 year output per person: 124,926.195

Some final takeaways from growth + Romer model

- The Romer model helps explain long term growth, unlike the Solow and Cobb Douglas models, because of the growth in nonrival ideas
- We can combine the Romer and Solow models to understand why and how countries reach a long run growth rate – we can still reach a “steady state” of capital, which affects short-run growth rates, especially with changes in our long-term parameters of savings and depreciation, but long run growth is driven by the generation of ideas
- These models aren't perfect and don't explain the complex political and economic relationships that have affected growth in different countries across the world, but they provide a foundation for us to understand the drivers of economic growth, in the short run and the long run

PSET 2, Q3

3. [10 points total] The Romer Model and the Zombie Apocalypse.

Braaaaaaaains! Consider an “old school” Romer economy² like the one we saw in class, with production of knowledge A and food Y . Scientists produce new knowledge using old knowledge and have productivity $\bar{z}$. Total population is fixed at $\bar{N}$ and employed either in knowledge or in food. The share of population in knowledge production equals ℓ , a number like 0.10.

$$\Delta A_t = \bar{z} A_t L_{at} \quad (3.1)$$

$$Y_t = A_t L_{yt} \quad (3.2)$$

$$\bar{N} = L_{at} + L_{yt} \quad (3.3)$$

$$L_{at} = \ell \bar{N} \quad (3.4)$$

Recall that the solution to the Romer Model reveals the following:

$$g[A] = \bar{z} \ell \bar{N} \quad (3.5)$$

$$A_t = A_0 (1 + g[A])^t \quad (3.6)$$

$$y_t = A_0 (1 + g[A])^t (1 - \ell) \quad (3.7)$$

Imagine what might happen if all the scientists accidentally infect themselves with the “zombie flu” and become zombies. Zombies:

- Do not produce new ideas
- Do not destroy old ideas
- Eat brains and do not eat food produced by farmers
- Do not eat farmers’ brains. They prefer tasty imported brains

PSET 2, Q3

- (a) [3 points] When the scientists become zombies, what happens to the growth rate of A ? What about the growth rate of y ?
- (b) [3 points] When the scientists become zombies, what happens to the level of A ? What about the level of output per person? (Consult the bullet points about zombies above.)

PSET 2, Q3

- (c) [4 points] Illustrate the Zombie Apocalypse in the graph below, showing any changes in the levels and growth rates in output per person over time. Discuss what you have drawn. Is the zombie flu good or bad? Explain.

PSET 2, Q3

(d) [4 points] Suppose rather than being turned into zombies, the scientists had all been deported to another country by the government. Would anything be different? Why?

(e) [4 points] Suppose rather than being turned into zombies, the scientists had all been repurposed as farmers by the government. Would anything be different? Why?